MIL-D-50677A (AR)
30 Nov 1981
SUPERSEDING
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15 April 1971

MILITARY SPECIFICATION

DETECTOR, DOPPLER: 10548509

This specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements, quality assurance provisions, and packaging for a doppler detector assembly used in Radar Receiver R-1475/VPS-2, part of Radar Set AN/VPS-2.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-13926	-	Fire Control Material: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	-	Parts, Equipment and Tools for Army Material: Packaging and Packing of
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-C-46357	-	Computer, Gun Direction; M18 Manufacturing Processes and Inspection of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-810 - Environmental Test Methods

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.
DRAWINGS (See 6.4)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

D10548509 - Detector, Doppler
D10548183 - Hybrid Junction

PACKAGING DATA SHEET

10548509 - Packaging of Doppler Detector

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Fabrication. The doppler detector shall be manufactured in accordance with Drawing D10548509 and drawings pertaining thereto.

3.2 General specification. Where applicable, the doppler detector shall meet the requirements of MIL-F-13926 and MIL-C-46357 as follows:

3.2.1 MIL-F-13926.

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishings

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d. Part identification and marking
e. Workmanship
f. Soldering electrical connections

3.2.2 MIL-C-46357.

a. Etched circuit assemblies
b. Wiring
c. Threaded fasteners
d. Marking
e. Finishes

3.3 Environmental service conditions.

3.3.1 Vibration. The doppler detector shall be mounted on the vibration table in normal service position. The doppler detector shall meet the requirements of 3.5 after being subject to continuous vibration for 2 hours along each of three mutually perpendicular axes within the frequency range of 10 to 55 hertz (Hz) and a half amplitude excursion of 1/64 inch. The rate of change of frequency at each frequency range shall not exceed 1 Hz per second. Vibration tests shall be conducted under ambient room conditions.

3.3.2 Shock. The doppler detector shall not suffer damage and shall meet the requirements of 3.5 after being subjected to 18 impact shocks, consisting of three shocks in opposite directions along each of the three mutually perpendicular axes. The shock impulses shall have an acceleration of 15 gravity units (G) and a pulse duration of 11 ±1 milliseconds (ms). The G value of each impulse shall be within ±10 percent when measured with a 0.2 to 250 Hz filter and the maximum G shall occur at approximately 5.5 ms.

3.3.3 Altitude.

3.3.3.1 Operating. The doppler detector shall meet the requirements of 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches Hg).

3.3.3.2 Nonoperating. The doppler detector shall meet the requirements of 3.5 after exposure for not less than 1 hour to barometric pressure encountered at 40,000 feet above sea level (5.5 inches Hg.)

3.3.4 Temperature.

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3.3.4.1 Operating. The doppler detector shall meet the requirements of 3.5 when stabilized thermally at any temperature between -40°F and +155°F.

3.3.4.2 Nonoperating (storage). The doppler detector shall not suffer permanent damage when subjected to not less than 4 hours at each temperature extreme of -65°F and +160°F, and shall meet the requirements of 3.5 when returned to room ambient.

3.3.5 Humidity (nonoperating). The doppler detector shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity of 95 ±5 percent for a period of not less than 10 days.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram of the doppler detector.

3.5 Performance.

3.5.1 When supplied with the following signals of a through d and modified as required in accordance with 3.5.1.1 through 3.5.1.4, the performance of 3.5.1.1 through 3.5.1.4 will be obtained:

a. Connect 25.0 ±0.1 volts direct current (Vdc) at 50 milliamperes (ma), with a shield to power supply case, to J1-1 (+), J1-2 (-) and J1-4(shield) of A1 and A2 of the doppler detector assembly.

b. Supply signals to A5J1 and A6J1 which are equal in amplitude, from a source which is variable between 59.93660 megahertz (MHz) and 59.94660 MHz in frequency and from -120 dBm to 0 dBm in amplitude, into a 50 ohm load.

c. Supply signals to A5J3 and A6J3 which are equal in amplitude, at a frequency of 59.93600 ±0.00020 MHz, at a level of +2.5 ±0.5 dBm into a 50 ohm load.

d. Load A1J1-A1 and A2J1-A1 with 150 ohms.

3.5.1.1 Noise. With the frequency of the signal applied to A5J1 and A6J1 set to 59.94000 ±.00020 MHz the signal measured at A1J1-A1 and A2J1-A1 shall be less than 31.0 millivolts (mV). The difference between the amplitudes measured at A1J1-A1 and A2J1-A1 shall not be greater than 3.0 dB.

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3.5.1.2 Gain Tracking.

a. When the signal level at A5J1 and A6J1 is increased to provide an output signal 5 dB above the greater noise level measured in 3.5.1.1, the difference in levels measured at A1J1-A1 and A2J1-A1 shall be less than 3 dB.

b. When the signal at A5J1 and A6J1 is increased to provide an output signal of 283.0 ±17.0 mV at the greater output of a, the input signal shall have increased by a minimum of 20 dB. The level difference at A1J1-A1 and A2J1-A1 shall be less than 2 dB.

3.5.1.3 Signal Tracking. When the signal level at A5J1 and A6J1 is set for -70 dBm, the phase difference between A1J1-A1 and A2J1-A1 shall be less than 10 degrees when the input frequency to A5J1 and A6J1 is varied from 59.93700 MHz to 59.94500 MHz (doppler frequency of 1000 Hz to 9,000 Hz).

3.5.1.4 S.C.V.

a. An input signal shall be applied to A5J1 with a frequency of 59.93600 ±0.00020 MHz, amplitude of -3.5 ±0.5 dBm into 50 ohms, switched ON and OFF at a 20 KHz rate with the switching rise and fall times at 40.0 ±5.0 nanoseconds (ns). The RMS output level at A1J1-A1 shall be noted.

b. When the frequency of the signal specified above is changed to 59.94000 ±0.00020 MHz the input level required to produce the same RMS output level as that noted in a, shall be at least 49.0 dB below the input level applied in a. The same requirement applies using A6J1 as the input and A2J1-A1 as the output.

3.6 First Article Inspection. This specification makes provisions for First Article Inspection. Requirements for the submission of First Article Samples by the contractor shall be as specified in the contract.

3.7 Workmanship. All parts and assemblies shall be free from burrs, chips, sharp edges, cracks, unblended radii, surface blisters, excess flash, corosion products, and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agents. Surface coating shall be continuous except for a few slight scratches not exposing the base material. The assembly operations shall be performed in a thorough, workmanlike manner consistent with the best practices of the industry. All required markings shall be neat and sharply defined.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract of purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified: the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First Article (initial production) submission. The requirement for first article approval and the responsibility (Government or contractor) for the first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three doppler detectors manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to ensure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certification of conformance for materials.

4.2.2 Contractor testing. When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certification for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

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4.3 Inspection provisions.

4.3.1 Submission of product. Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies. All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provision (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.3.2.2 Final acceptance inspection. Subsequent to the first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in 4.3.2.3 and 4.4 herein. Examination and test for packaging and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in 4.3.2.3 and 4.4. shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

4.3.2.3 Functional tests. The requirements and test in Table I shall be inspected on a 100 percent basis. All examinations and tests shall be conducted at the standard ambient temperature (60°F to 90°F).

TABLE I. Classification of Defects
(100 percent inspection)

Characteristic	Requirement	Test Procedures
Critical: None		
Major:		
101. Electrical	3.4	4.6.2
102. Performance	3.5	4.6.3
Minor: None		

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4.3.3 Acceptance and rejection. Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling. One doppler detector shall be selected at random by a Government representative as a control sample from each 100 produced. The control sample shall meet the requirements and tests in Table II. Three doppler detectors shall be selected at random by a Government representative as a control sample from each 50 produced or from each month's production, whichever occurs first. The control samples shall meet the requirements and tests in Table III.

TABLE II. Special Sampling

Control Test	Requirement	Test Procedures
301. Fabrication	3.1	Applicable drawings - Visual
302. General Specifications	3.2	MIL-F-13926 & MIL-C-46357 - Visual

TABLE III. Special Sampling, Environmental

Characteristic	Requirement	Test Procedures
303. Vibration	3.3.1	4.6.1.1
304. Shock	3.3.2	4.6.1.2
305. Altitude	3.3.3	4.6.1.3
306. Temperature	3.3.4	4.6.1.4
307. Humidity (nonoperating)	3.3.5	4.6.1.5

4.4.1 Failure of sample. Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by Government until necessary corrections have been made by the contractor and a resubmitted sample has been approved (see 4.3.3).

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4.5 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.5.2 Contractor-furnished inspection equipment.

4.5.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specification or SQAP forming a part of the contract shall be supplied by the contractor in accordance with applicable technical data listed in the TDP.

4.5.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Test equipment. In conjuction with 4.5.2.2, the standard test equipment listed in Table IV shall be utilized in performing the applicable test of 4.6.

4.6 Test methods and procedures.

4.6.1 Environmental. Environmental tests shall be conducted in accordance with the general requirements and test methods of MIL-STD-810 except as specified herein. Testing equipment used in environmental tests shall be in accordance with "Test Facilities and Apparatus" requirements of MIL-STD-810.

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4.6.1.1 Vibration. The test shall be performed as specified in Method 514, Procedure VIII, of MIL-STD-810 as modified by paragraph 3.3.1 herein. After vibration, the doppler detector shall be tested as specified in 4.6.3 to determine compliance with 3.3.1.

4.6.1.2 Shock. This test shall be performed as specified in Procedure I, Method 516 of MIL-STD-810 except that shock pulses shall be 11.0 ms and 15 G. After the shock impulses, the doppler detector shall be tested as specified in 4.6.3 to determine compliance with 3.3.2.

4.6.1.3 Altitude (nonoperating and operating). This test shall be performed as specified in Procedure I, Method 500 of MIL-STD-810 except that the operating altitude shall be limited to 10,000 feet (20.6 inches of Hg). Operation shall be the performance tests of 4.6.3 herein. Nonoperating test at 40,000 feet (5.5 inches Hg) shall also be performed.

TABLE IV. Test Equipment List

Item	Description
1. RF Signal generator (2)	Frequency: 60 MHz variable ±5 KHz Stability: 250 Hz after 15 min. warm-up Output: to +3 dBm into 50 ohms
2. Phase Meter	Frequency: 600 Hz to 10 KHz min. Input signal range: 0.2 mV to 1 V RMS min. Display: ±18°min. Resolution: 0.1° Phase Accuracy: ±0.5° Input Impedance: 1 Meg ohm min.
3. Frequency Counter	Frequency: 60 MHz Resolution: 25 Hz Sensitivity: 50 mV Input Impedance: 50 ohm
4. RMS Voltmeter	Range: 300 mV full scale Frequency: 10 Hz to 10 KHz minimum Input Impedance: 1 meg ohm minimum Accuracy: ±1.0%

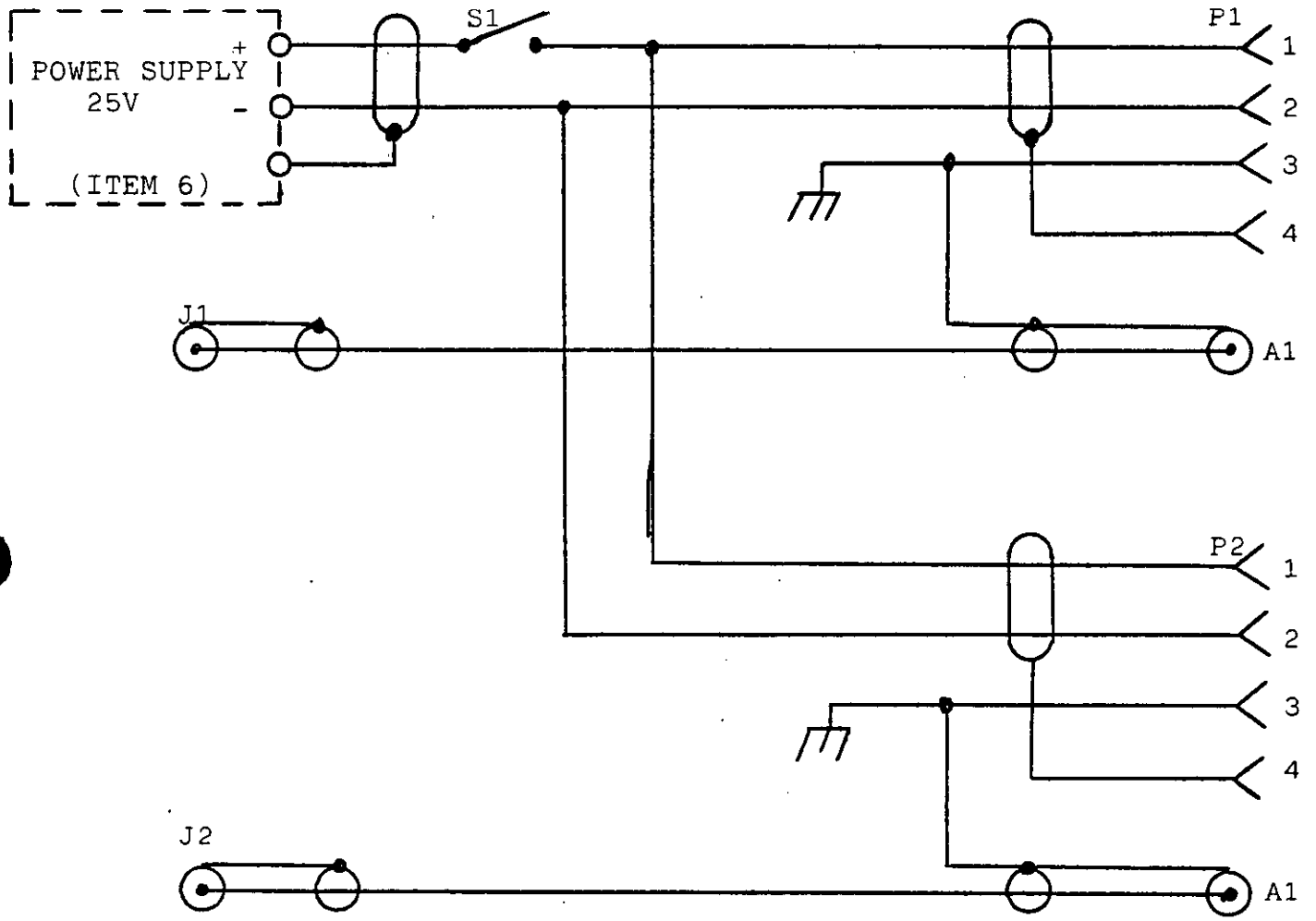
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5.	Variable Attenuator (2)	Range: 0 to 120 dB in 1 dB steps Impedance: 50 ohms Accuracy: ±0.5 dB of setting
6.	Power Supply	Output: 24 to 26 volts at 50 ma Ripple: less than 5 mV
7.	Oscilloscope	Frequency response: 1 MHz Time Base: 1 us to 100 ms/cm Sensitivity: 20 mV to 1 V/cm Vertical amplifier: dual channel Modes: chan 1, chan 2, chopped, add, AC/DC coupled
8.	Digital DC Voltmeter	Range: 10-30 volts Accuracy: ±0.1%
9.	R.F. Voltmeter	Range: 300 mV full scale Frequency: 40 MHz to 80 MHz Accuracy: ±2% Input Impedance: 50 ohms
10.	Pulse Generator	Repetition note: 20.0 ±2.0 KHz Pulse Width: 0.4 ±0.02 us Amplitude: +0.50 ±0.02 V Rise and Fall time: 40.0 ±5.0 ns
11.	Test Harness	See Figure 1
12.	Hybrid Junction	Frequency: 60 MHz Outputs: Two (2) Coupling: -3 dB Isolation: 30 dB min Amplitude Balance: 0.2 dB max. Insertion loss: 0.75 dB max. Impedance: 50 ohms VSWR: 1.3:1 max.
13.	Termination (2)	150 ohm
14.	BNC Tee (2)	UG-274/U

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15.	Power Divider	Frequency: 60 MHz Outputs: Four (4) Coupling: -6 dB Isolation: 30 dB min. Amplitude Balance: 0.2 dB max. Insertion Loss: 0.7 dB Impedance: 50 ohms VSWR: 1.3:1 max.
16.	Termination (2)	50 ohm
17.	Double balanced mixer	Frequency: LO 60 MHz RF 60 MHz Impedance: 50 ohms
18.	Phase Shifter	Frequency: 60 MHz Bandwidth: 10% Phase Shift Range: 0-180 min. VSWR: 1.5:1 max. Impedance: 50 ohms Insertion loss: 1dB max.

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NOTES: 1. P1 and P2 = ITT Cannon or Cinch P/N DEM-5WIS.
2. J1 and J2 = BNC Male, UG-88/U or equivalent.
3. S1 = Cutler - Hammer 7580K7 or equivalent.

FIGURE 1. Test Harness

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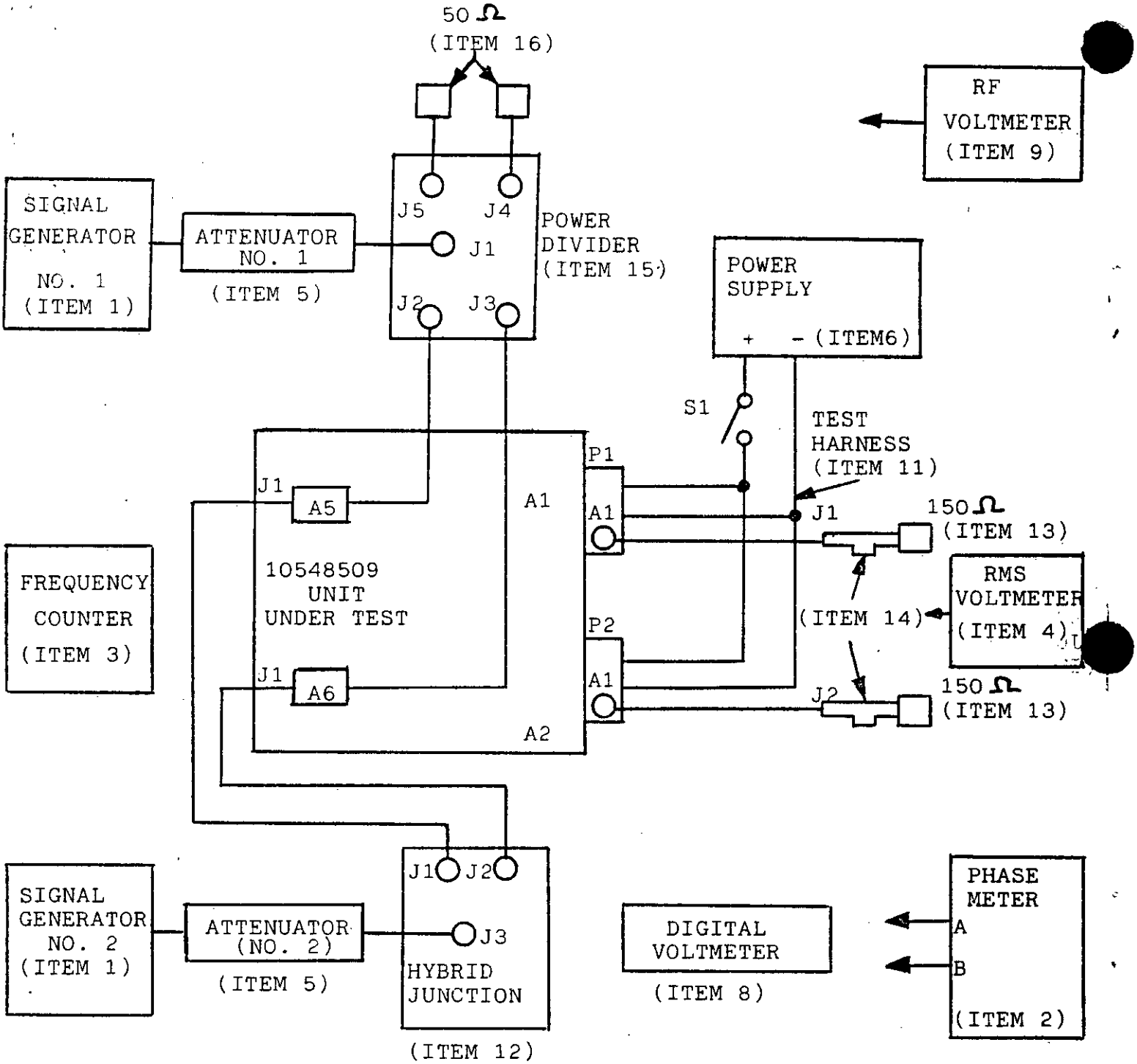


FIGURE 2. Test Setup
Noise, Gain and Signal Tracking

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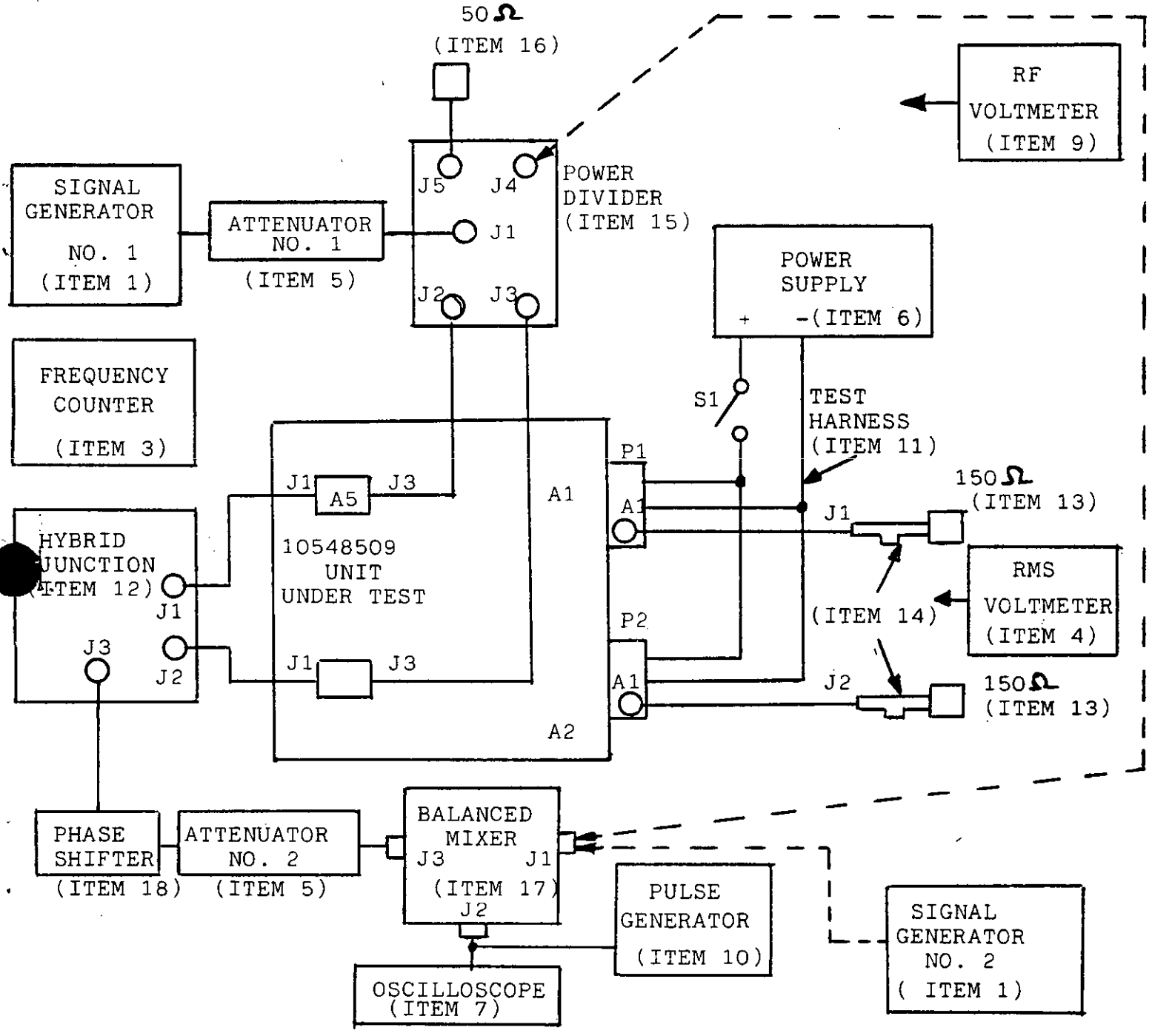


FIGURE 3. Test Setup S.C.V.

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4.6.1.4 Temperature

4.6.1.4.1 High (Nonoperating and Operating) This test shall be performed as specified in Procedure I, Method 501 of MIL-STD-810. The temperature of Step 2 shall be +160°F. The temperature of Step 3 shall be maintained for not less than 4 hours. The temperature of Step 4 shall be +155°F. In Step 5, the temperature shall be maintained for 4 hours and the doppler detector shall be tested in accordance with 4.6.3 during this time.

4.6.1.4.2 Low (Nonoperating and Operating). This test shall be performed as specified in Procedure I, Method 502 of MIL-STD-810. The temperature of Step 2 shall be -65°F maintained for not less than 4 hours. The temperature of Step 4 shall be -40°F. In Steps 5 and 7, the doppler detector shall be tested in accordance with 4.6.3.

4.6.1.5 Humidity (Nonoperating). This test shall be performed in accordance with Procedure II, Method 507 of MIL-STD-810. The doppler detector shall be tested in accordance with 4.6.3.

4.6.2 Electrical. The doppler detector shall be inspected for continuity using standard measuring equipment (SME) and visual inspection for compliance with 3.4.

4.6.3 Performance. The doppler detector shall be tested as follows:

a. Connect the equipment as shown in Figure 2. Set PWR switch on test harness, Item 11, to OFF. The test equipment shall have a 15 minute warmup. The two RF signal generators, Item 1, are monitored with frequency counters, Item 3, until the signals stabilize to allow a test to be completed with less than 250 Hz total relative drift.

b. Set Attenuator #1, Item 5 to 3 dB. Connect RF voltmeter, Item 9, to J4 of the Power Divider, Item 15, and adjust Signal Generator #1, Item 1, for 300.0 ±5.0 mV read on the RF Voltmeter, Item 9, at a frequency of 59.93600 ± .00020 MHz as read on the frequency counter.

c. Set Signal Generator #2, Item 1, output level for -17 dBm as indicated on the signal generator meter, and a frequency of 59.9400 ± 0.00020 MHz as indicated on the frequency counter. Set Attenuator #2, Item 5, to 101 dB.

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d. Set Power Supply, Item 6, for 25.0 ± 0.1 Vdc as measured with the digital voltmeter, Item 8.

4.6.3.1 Noise. With the equipment set up as in 4.6.3 a, b, c, and d, turn PWR switch on the test harness, Item 11, to ON. Connect J1 (loaded with 150 ohms, Item 13) to the RMS Voltmeter, Item 4, and record the output of amplifier A1. LIMIT: 31.0 mV maximum. Disconnect J1 and connect J2 to the RMS Voltmeter (loaded with 150 ohms) and record the output of amplifier A2. LIMIT: 31.0 mV maximum. Calculate the difference between the two readings. LIMIT: 3.0 dB maximum.

4.6.3.2 Gain Tracking. Connect the equipment so the larger noise level of 4.6.3.1 is being monitored on the RMS voltmeter. Reduce attenuator #2, Item 5, for an output reading 5 dB higher than the noise level measured in 4.6.3.1. Record the setting of attenuator #2. Record the level at both J1 and J2 and calculate the difference. LIMIT: 3.0 dB maximum.

4.6.3.2.1 Connect the larger signal of J1 and J2 to the RMS voltmeter and reduce attenuator #2 for an output reading of 283.0 ± 17.0 mV. Record the setting of attenuator #2. It shall be at least 20 dB less than the setting of 4.6.3.2. Record the output level at both J1 and J2 and calculate the difference. LIMIT: 2.0 dB maximum.

4.6.3.3 Signal Tracking. Connect the equipment as shown in Figure 2. Set signal generator #1 and attenuator #1 as specified in 4.6.3.b. Set signal generator #2 frequency for 59.93660 ± 0.00020 MHz with an output level of -10 dBm. Set attenuator #2 for 60 dB of attenuation. Connect J1 and J2 (loaded with 150 ohms) to the A and B inputs of the phase meter, Item 2.

4.6.3.3.1 Vary signal generator #2 to each of the frequencies listed below and record the phase angle reading at each frequency. Subtract the lowest phase angle reading from the highest. The difference shall not exceed 10 degrees.

FREQUENCY (+0.00020 MHz)

	59.94100 MHz
59.93700 MHz	59.94200 MHz
59.93800 MHz	59.94300 MHz
59.93900 MHz	59.94400 MHz
59.94000 MHz	59.94500 MHz

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4.6.3.4 S.C.V. Connect equipment as shown in Figure 3. Set the pulse generator, Item 10, as follows:

Repetition Rate	20.0 ± 2.0 KHz
Pulse Width	0.40 ± 0.02 us
Amplitude	+ 0.50 ± 0.02 V
Rise Time	-40.0 ± 5.0 ns
Fall Time	-40.0 ± 5.0 ns

Set signal generator #1 and attenuator #1 as specified in 4.6.3.b. Set signal generator #2 to 59.94000 ± 0.00020 MHz with an output level of 300.0 ± 5.0 mV as measured with the RF Voltmeter, Item 9.

4.6.3.4.1 Connect J4 of power divider, Item 15, to J1 of the balanced mixer, Item 17. Adjust attenuator #2 for 0 dB attenuation. Adjust the phase shifter, Item 18, for a peak J1 indication on the RMS voltmeter, Item 4. Record the RMS voltmeter reading.

4.6.3.4.2 Set attenuator #2 for 60 dB of attenuation. Disconnect the power divider J4 from J1 of the balanced mixer and connect the signal generator #2 output to J1 of the balanced mixer. Adjust attenuator #2 for the same RMS voltmeter reading of 4.6.3.4.1. Record the attenuator #2 total attenuation. LIMIT: 49.0 dB minimum.

4.6.3.4.3 Connect the equipment so J2 output is being monitored on the RMS voltmeter. Repeat steps 4.6.3.4.1 and 4.6.3.4.2 for A2.

5. PACKAGING

5.1 Packaging, packing, and marking. Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 10548509. The levels of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. The doppler detector is used to remove any doppler effect from the i-f frequency of radar return signals in Radar Set AN/VPS-2. This doppler is caused by the transmitted rf frequency of the radar striking a moving target and the return frequency being shifted in frequency. This frequency shift is used to obtain an indication of moving targets.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Applicable packaging data sheet number (see 5.1).

c. Selection of applicable level of preservation packaging and packing.

d. Level of protection required.

e. Applicable stock number.

f. Requirement for first article submission.

6.3 Submission of designs for approval. Contractor designs shall be approved by the Government prior to fabricating or procuring of the equipment. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. Submit designs as required to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAF-I, Dover, NJ 07801.

When the contractor submits inspection equipment designs to the Government for approval, he shall give the following information in his letter of transmittal:

a. The contract number.
b. The contract item (name, model number, etc.)
c. The designs remaining to be submitted and the expected date of transmittal.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

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6.5 Changes to previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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Army-AR

Preparing activity:
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